The Bellman Policy-Evaluation Operator L_d: Norms, Contraction, and Fixed Point Iteration

Reinforcement Learning

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Contents

1 Infinity Norm and Induced Operator Norm **2**
1.1 Infinity Norm on Vectors 2
1.2 Induced Operator Norm 2
1.3 Theorem: Induced Infinity Norm Formula 2

2 Bellman Affine Operator on the Space of Bounded Functions **4**
2.1 The Space V 4
2.2 Induced Reward and Transition Kernel 4
2.3 Lemma: Well-Definedness of the Bellman Affine Map 4

3 Theorem 6.2.3 (Banach Fixed-Point Theorem) **5**

4 Iterative Policy Evaluation as a Fixed-Point Method **7**
4.1 Contraction of the Bellman affine Operator 8
4.2 Iterative Policy Evaluation Algorithm 8

1 Infinity Norm and Induced Operator Norm

1.1 Infinity Norm on Vectors

Let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$.

The **infinity norm** (sup norm) is defined as

$$\|x\|_\infty := \sup_{1 \le j \le n} |x_j|.$$

Interpretation.

- Measures the largest absolute component of the vector.
- The unit ball is a cube in $\mathbb{R}^n$.
- Captures worst-case coordinate magnitude.

1.2 Induced Operator Norm

Let

$$A : \mathbb{R}^n \to \mathbb{R}^m$$

be a linear operator represented by a matrix

$$A = (a_{ij}) \in \mathbb{R}^{m \times n}.$$

The operator norm induced by the infinity vector norm is defined as

$$\|A\|_\infty := \sup_{x \neq 0} \frac{\|Ax\|_\infty}{\|x\|_\infty} = \sup_{\|x\|_\infty = 1} \|Ax\|_\infty.$$

Interpretation. The operator norm measures the largest possible amplification of the transformation.

1.3 Theorem: Induced Infinity Norm Formula

Theorem 1.1. *Let $A \in \mathbb{R}^{m \times n}$. Then*

$$\boxed{\|A\|_\infty = \sup_{1 \le i \le m} \sum_{j=1}^{n} |a_{ij}|}$$

That is, the induced infinity operator norm equals the supremum of the absolute row sums.

Proof. **Step 1: Upper bound.**

For any $x \neq 0$,

$$(Ax)_i = \sum_{j=1}^{n} a_{ij} x_j.$$

Hence,

$$|(Ax)_i| \leq \sum_{j=1}^{n} |a_{ij}||x_j| \leq \sum_{j=1}^{n} |a_{ij}| \, \|x\|_\infty.$$

Taking supremum over i,

$$\|Ax\|_\infty = \sup_i |(Ax)_i| \leq \left(\sup_i \sum_{j=1}^{n} |a_{ij}| \right) \|x\|_\infty.$$

Dividing by $\|x\|_\infty$ and taking supremum over x,

$$\|A\|_\infty \leq \sup_{1 \leq i \leq m} \sum_{j=1}^{n} |a_{ij}|.$$

Step 2: Lower bound.

Let

$$R(A) := \sup_{1 \leq i \leq m} \sum_{j=1}^{n} |a_{ij}|.$$

For any $\varepsilon > 0$, there exists i_ε such that

$$\sum_{j=1}^{n} |a_{i_\varepsilon j}| > R(A) - \varepsilon.$$

Define $x^{(\varepsilon)} \in \mathbb{R}^n$ by

$$x_j^{(\varepsilon)} := \begin{cases} \operatorname{sign}(a_{i_\varepsilon j}) & \text{if } a_{i_\varepsilon j} \neq 0, \\ 1 & \text{otherwise.} \end{cases}$$

Then $\|x^{(\varepsilon)}\|_\infty = 1$, and

$$(Ax^{(\varepsilon)})_{i_\varepsilon} = \sum_{j=1}^{n} |a_{i_\varepsilon j}|.$$

Thus,

$$\|Ax^{(\varepsilon)}\|_\infty \geq R(A) - \varepsilon.$$

Since ε is arbitrary,

$$\|A\|_\infty \geq R(A).$$

Combining both inequalities gives the result. $\square$

2 Bellman Affine Operator on the Space of Bounded Functions

2.1 The Space V

Let S be finite or countable and define

$$V := \left\{ v : S \to \mathbb{R} \ : \ \|v\|_\infty := \sup_{s \in S} |v(s)| < \infty \right\}.$$

V is a Banach space under the infinity norm.

2.2 Induced Reward and Transition Kernel

Fix a stationary randomized decision rule d.

Define:

$$r_d(s) := \sum_{a \in A_s} q_d(s,a)\, r(s,a),$$

$$p_d(j|s) := \sum_{a \in A_s} q_d(s,a)\, p(j|s,a).$$

Let P_d denote the corresponding transition operator.

2.3 Lemma: Well-Definedness of the Bellman Affine Map

Lemma 2.1. *Assume* $|r(s,a)| \le M$ *and* $0 \le \lambda < 1$.

Define

$$(L_d v)(s) := r_d(s) + \lambda (P_d v)(s).$$

Then

$$L_d : V \to V.$$

Proof. **Step 1:** $r_d \in V$**.**

$$|r_d(s)| \le \sum_{a \in A_s} q_d(s,a)\, |r(s,a)| \le M.$$

Thus $\|r_d\|_\infty \le M$.

Step 2: $\|P_d v\|_\infty \le \|v\|_\infty$**.**

Since $p_d(\cdot|s)$ is a probability distribution,

$$|(P_d v)(s)| \le \sum_j p_d(j|s)\, |v(j)| \le \|v\|_\infty.$$

Equivalently, since P_d is row-stochastic,

$$\|P_d\|_\infty = 1,$$

and hence

$$\|P_d v\|_\infty \le \|P_d\|_\infty \|v\|_\infty = \|v\|_\infty.$$

Step 3: Combine.

$$\|L_d v\|_\infty \le \|r_d\|_\infty + \lambda \|P_d v\|_\infty \le M + \lambda \|v\|_\infty < \infty.$$

Thus $L_d v \in V$. □

Remark. L_d is **affine**, not linear, since

$$L_d(0) = r_d \neq 0$$

unless rewards are identically zero.

This lemma establishes closure of the Bellman affine operator on V, which is required before proving contraction and applying the Banach fixed-point theorem.

3 Theorem 6.2.3 (Banach Fixed-Point Theorem)

Theorem 3.1 (Theorem 6.2.3 (Banach Fixed-Point Theorem))**.** *Suppose V is a Banach space and $T : V \to V$ is a contraction mapping. That is, there exists λ with $0 \le \lambda < 1$ such that*

$$\|Tv - Tu\| \le \lambda \|v - u\| \quad \textit{for all } u, v \in V.$$

Then:

(a) There exists a unique $v^* \in V$ such that $Tv^* = v^*$.

(b) For arbitrary $v^0 \in V$, the sequence $\{v^n\}$ defined by

$$v^{n+1} = Tv^n = T^{n+1} v^0$$

converges to v^*.

Proof. **Intuition (what we are trying to do).** A *Banach space* is a complete normed space, meaning that every Cauchy sequence converges to a limit *inside* V.

So the proof has a forced structure:

- Show the iteration $\{v^n\}$ is Cauchy.
- Use completeness to conclude it converges to some $v^* \in V$.
- Show the limit point must satisfy $Tv^* = v^*$.
- Show there cannot be two different fixed points.

A *contraction* is exactly the property that makes successive iterates get closer by a uniform factor $\lambda < 1$.

Step 1: Show that $\{v^n\}$ is Cauchy.

Step 1(a): Successive differences shrink geometrically. From $v^{n+1} = Tv^n$,

$$\|v^{n+1} - v^n\| = \|Tv^n - Tv^{n-1}\| \leq \lambda \|v^n - v^{n-1}\|.$$

Applying the same inequality repeatedly gives

$$\boxed{\|v^{n+1} - v^n\| \leq \lambda^n \|v^1 - v^0\| \quad \text{for all } n \geq 0.}$$

Intuition. Each application of T shrinks distances by at least a factor $\lambda < 1$, so the step sizes decay like λ^n.

Step 1(b): Use the Cauchy definition via triangle inequality. Let $m \geq 1$. Insert intermediate points:

$$\|v^{n+m} - v^n\| \leq \sum_{k=n}^{n+m-1} \|v^{k+1} - v^k\|.$$

Using the geometric bound from Step 1(a),

$$\|v^{n+m} - v^n\| \leq \sum_{k=n}^{n+m-1} \lambda^k \|v^1 - v^0\|.$$

Compute the geometric sum:

$$\sum_{k=n}^{n+m-1} \lambda^k = \lambda^n \frac{1 - \lambda^m}{1 - \lambda} \leq \frac{\lambda^n}{1 - \lambda}.$$

Hence,

$$\boxed{\|v^{n+m} - v^n\| \leq \frac{\lambda^n}{1 - \lambda} \|v^1 - v^0\|.}$$

Since $0 \leq \lambda < 1$, we have $\lambda^n \to 0$, so for any $\varepsilon > 0$ we can choose N large enough that for all $n \geq N$ and all $m \geq 1$,

$$\|v^{n+m} - v^n\| < \varepsilon.$$

Thus $\{v^n\}$ is Cauchy.

Intuition. Far out in the sequence, the total jump between two iterates is bounded by the tail of a geometric series, so it must become arbitrarily small.

Step 2: Use completeness of V to obtain a limit point. Because V is Banach (complete), every Cauchy sequence converges in V. Therefore there exists $v^* \in V$ such that

$$\lim_{n \to \infty} v^n = v^*.$$

Intuition. The contraction makes the iterates bunch up tighter and tighter; completeness guarantees there is an actual point in V where they land.

Step 3: Show $Tv^* = v^*$ (two equivalent proofs).

Step 3(a): Limit-interchange proof (using continuity of T). Since T is a contraction, it is Lipschitz:

$$\|Tv - Tu\| \leq \lambda \|v - u\|.$$

Hence T is continuous. Therefore,

$$\lim_{n\to\infty} Tv^n = T\left(\lim_{n\to\infty} v^n\right) = Tv^*.$$

But $Tv^n = v^{n+1}$, so

$$\lim_{n\to\infty} Tv^n = \lim_{n\to\infty} v^{n+1} = v^*.$$

Hence $Tv^* = v^*$.

Step 3(b): Norm-trick proof (no interchange needed). Consider

$$\|Tv^* - v^*\|.$$

Add and subtract Tv^n and apply triangle inequality:

$$\|Tv^* - v^*\| \leq \|Tv^* - Tv^n\| + \|Tv^n - v^*\|.$$

Use contraction on the first term and $Tv^n = v^{n+1}$ on the second term:

$$\|Tv^* - v^*\| \leq \lambda \|v^* - v^n\| + \|v^{n+1} - v^*\|.$$

Let $n \to \infty$. Since $v^n \to v^*$, both terms go to 0, so $\|Tv^* - v^*\| = 0$ and thus $Tv^* = v^*$.

Intuition. Once v^n is very close to v^*, applying T cannot separate them much (contraction), and also $Tv^n = v^{n+1}$ is close to v^*. So Tv^* is forced to equal v^*.

Step 4: Uniqueness of the fixed point (clean). Suppose $u^* \in V$ is another fixed point, so $Tu^* = u^*$. Then

$$\|v^* - u^*\| = \|Tv^* - Tu^*\| \leq \lambda \|v^* - u^*\|.$$

Subtract the right-hand side:

$$(1 - \lambda)\|v^* - u^*\| \leq 0.$$

Since $1 - \lambda > 0$ and norms are nonnegative, this forces $\|v^* - u^*\| = 0$, hence $v^* = u^*$.

Intuition. A contraction must strictly shrink the distance between two distinct points. Two fixed points would not move at all, so their distance could not shrink. Only possible if the distance is already zero.

This completes the proof. □

4 Iterative Policy Evaluation as a Fixed-Point Method

We have shown that for a stationary Markov decision rule $d \in D^{MR}$, the discounted value function $v_\lambda^{d^\infty}$ satisfies the fixed-point equation

$$v = L_d v, \qquad (L_d v)(s) := r_d(s) + \lambda (P_d v)(s).$$

We now show that this fixed point can be computed by iteration because L_d is a contraction mapping on the Banach space $(V, \|\cdot\|_\infty)$.

4.1 Contraction of the Bellman affine Operator

Lemma 4.1 (Contraction of the Bellman affine operator)**.** *Let $0 \le \lambda < 1$. Then for all $v, w \in V$,*

$$\|L_d v - L_d w\|_\infty \le \lambda \|v - w\|_\infty.$$

Hence $L_d : V \to V$ is a contraction mapping with modulus λ.

Proof. For any $s \in S$,

$$(L_d v)(s) - (L_d w)(s) = \big(r_d(s) + \lambda (P_d v)(s)\big) - \big(r_d(s) + \lambda (P_d w)(s)\big) = \lambda (P_d(v - w))(s).$$

Taking the sup norm,

$$\|L_d v - L_d w\|_\infty = \lambda \|P_d(v - w)\|_\infty.$$

Since P_d is stochastic (each row is a probability distribution), it does not increase the ∞-norm:

$$\|P_d u\|_\infty \le \|u\|_\infty \qquad \forall u \in V.$$

Therefore,

$$\|L_d v - L_d w\|_\infty \le \lambda \|v - w\|_\infty.$$

□

Corollary 4.1.1 (Existence, uniqueness, and convergence of the iteration)**.** *Because $(V, \|\cdot\|_\infty)$ is Banach and L_d is a contraction, Theorem 6.2.3 (Banach fixed-point theorem) implies:*

- *there exists a unique $v^* \in V$ such that $L_d v^* = v^*$;*
- *for any initialization $v^{(0)} \in V$, the iteration*

  $$v^{(k+1)} = L_d v^{(k)} = r_d + \lambda P_d v^{(k)}$$

  converges to v^*.

Moreover, this unique fixed point is exactly the policy value function:

$$v^* = v_\lambda^{d^\infty}.$$

4.2 Iterative Policy Evaluation Algorithm

Algorithm 1: Iterative Policy Evaluation (Fixed-Point Iteration for $v = L_d v$)

Input: Stationary Markov decision rule d, induced reward r_d, induced transition matrix P_d, discount $\lambda \in (0, 1)$, tolerance $\varepsilon > 0$.
Output: Approximation v to $v_\lambda^{d^\infty}$.
Initialize $v^{(0)} \in V$ (e.g., $v^{(0)} \equiv 0$);
for $k = 0, 1, 2, \ldots$ **do**
  $v^{(k+1)} \leftarrow r_d + \lambda P_d v^{(k)}$;
  if $\|v^{(k+1)} - v^{(k)}\|_\infty < \varepsilon$ **then**
    return $v^{(k+1)}$;